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**PROJECT
Of
WEB DESIGNING
Subject Code: 25CAH-153**

TASK MANAGER

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1. Introduction

Task Manager is a web-based application developed using core front-end technologies — HTML, CSS, and JavaScript. The application enables users to efficiently organize, track, and manage their daily tasks in a structured and user-friendly interface.

In today's fast-paced world, managing personal and professional tasks effectively has become essential. This project addresses that need by providing a lightweight, browser-based task management solution that requires no external backend or database, relying instead on the browser's `localStorage` for persistent data storage.

The application allows users to create tasks with essential metadata such as task name, details, due date, importance level (High/Medium/Low), and difficulty level (Hard/Medium/Easy). Each task can be assigned a status — Pending, In Progress, or Completed — and can be filtered and sorted dynamically. The interface also provides real-time statistics and visual indicators for overdue tasks.

2. Objectives

The key objectives of this project are:

- To build a fully functional task manager using only HTML, CSS, and JavaScript without any external frameworks.
- To allow users to add, view, update, and delete tasks seamlessly.
- To provide task categorization based on importance (High, Medium, Low) and difficulty (Hard, Medium, Easy).
- To implement persistent data storage using browser `localStorage` so tasks survive page reloads.
- To display real-time statistics showing the count of Total, Pending, In Progress, and Done tasks.
- To provide filtering by task status and sorting by due date, importance, or difficulty.
- To visually indicate overdue and today-due tasks with color-coded labels.
- To deliver a clean, responsive, and intuitive user interface for all screen sizes.

3. Data Description

Each task stored in the application contains the following data fields:

Field	Type / Values	Description
id	Number (timestamp)	Unique identifier generated via Date.now()
name	String	Name/title of the task
details	String	Optional description or extra information
date	String (YYYY-MM-DD)	Due date of the task
importance	high / medium / low	Priority level of the task
difficulty	hard / medium / easy	Complexity level of the task
status	pending / in-progress / completed	Current state of the task

Data is stored in the browser's localStorage under the key 'tasks_basic' as a JSON array. The data persists across sessions until the user clears browser storage.

4. Tools and Technologies

The following tools and technologies were used to develop this project:

Technology	Purpose / Usage
HTML5	Page structure, forms, input elements, layout semantics
CSS3	Styling, layout (Flexbox), color-coded priorities, responsive design
JavaScript (ES5)	DOM manipulation, task logic, localStorage, dynamic rendering
localStorage API	Persistent client-side storage of task data as JSON
VS Code	Code editor used for development
Google Chrome	Browser used for testing and output verification
GitHub	Version control and project hosting

4.1 Code with Explanation

The project is developed as a single HTML file. Below is a breakdown of each section:

HTML Structure:

The HTML section defines the visual layout. It includes a statistics bar showing task counts, an input form for adding tasks, filter and sort controls, and a task list container. All elements are identified by unique IDs for JavaScript access.

CSS Styling:

The CSS applies a clean, minimal design. The task cards use colored left borders (red for High, orange for Medium, green for Low importance). Flexbox is used throughout for responsive layouts. Completed tasks receive reduced opacity and strikethrough text. Overdue tasks are highlighted in red and today-due tasks in orange.

JavaScript Functions:

Function	Description
<code>save()</code>	Serializes the tasks array to JSON and saves it in localStorage.
<code>addTask()</code>	Reads form inputs, validates them, creates a task object, pushes to array, saves and re-renders.
<code>deleteTask(id)</code>	Filters out the task with the matching ID after user confirmation, saves and re-renders.
<code>updateStatus(id, status)</code>	Finds a task by ID and updates its status property, then saves and re-renders.
<code>dueLabel(date, status)</code>	Returns an HTML string with color-coded labels: Overdue, Due today, Due tomorrow, or formatted date.
<code>render()</code>	Applies active filter and sort, builds HTML for all tasks, injects into DOM, and updates the stats bar.

Key Code Snippet — addTask() Function:

```
function addTask() {  
  
    var name = document.getElementById('name').value.trim();
```

```
var date = document.getElementById('date').value;

if (!name) { alert('Please enter a task name.');
```



```
return; }

tasks.push({ id: Date.now(), name, date, ... status: 'pending' });

save(); render();

}
```

4.2 Output

The following screenshot shows the Task Manager application running in a web browser. The interface displays the statistics bar at the top, followed by the task input form, filter/sort controls, and the task list area.

The screenshot displays the 'Task Manager' application interface. At the top, the title 'Task Manager' is centered. Below it is a statistics bar with four boxes, each showing a count of 0: 'Total', 'Pending', 'In Progress', and 'Done'. The main form area contains several input fields: 'Task Name' (placeholder: 'What needs to be done?'), 'Details' (placeholder: 'Any extra info...'), 'Due Date' (placeholder: 'dd-mm-yyyy'), 'Importance' (dropdown menu set to 'Medium'), and 'Difficulty' (dropdown menu set to 'Medium'). A black 'Add Task' button is positioned below these fields. At the bottom of the form area, there are two dropdown menus: 'All Tasks' and 'Sort by...'. The task list area at the very bottom is empty and contains the text 'No tasks here.'

Figure 1: Task Manager Application - Output Screenshot

5. Conclusion

This project successfully demonstrates the development of a fully functional task management application using only front-end technologies — HTML, CSS, and JavaScript. The application meets all the defined objectives, providing a practical and user-friendly interface for creating, tracking, and managing tasks.

The use of `localStorage` for data persistence eliminates the need for a backend server, making the application lightweight and easy to deploy in any web browser. The color-coded priority system, real-time statistics, due-date indicators, and dynamic filtering and sorting features collectively make it an effective productivity tool.

The project highlights how fundamental web technologies can be combined effectively to build a complete, real-world application without reliance on external libraries or frameworks. It also demonstrates important programming concepts such as DOM manipulation, event handling, JSON serialization, and client-side storage.

6. Learning Outcomes

Through the development of this project, the following learning outcomes were achieved:

- Gained hands-on experience with HTML5 semantic elements, forms, and input validation.
- Developed proficiency in CSS3 including Flexbox layout, pseudo-classes, and responsive design patterns.
- Strengthened JavaScript skills including DOM manipulation, event handling, array methods (filter, find, sort), and JSON operations.
- Learned how to use the Web Storage API (`localStorage`) for persistent client-side data management.
- Understood how to build a complete CRUD (Create, Read, Update, Delete) application in vanilla JavaScript.
- Practiced writing clean, modular, and maintainable front-end code.
- Developed understanding of UX principles such as color coding for status, visual feedback for user actions, and responsive layouts.
- Gained experience using Git and GitHub for version control and project management.

7. GitHub Link

The complete source code for this project is hosted on GitHub. The repository contains the full project files including the HTML, CSS, and JavaScript code.

GitHub Repository: